

2023 AMC 10B Problem 19 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 19

Problem:

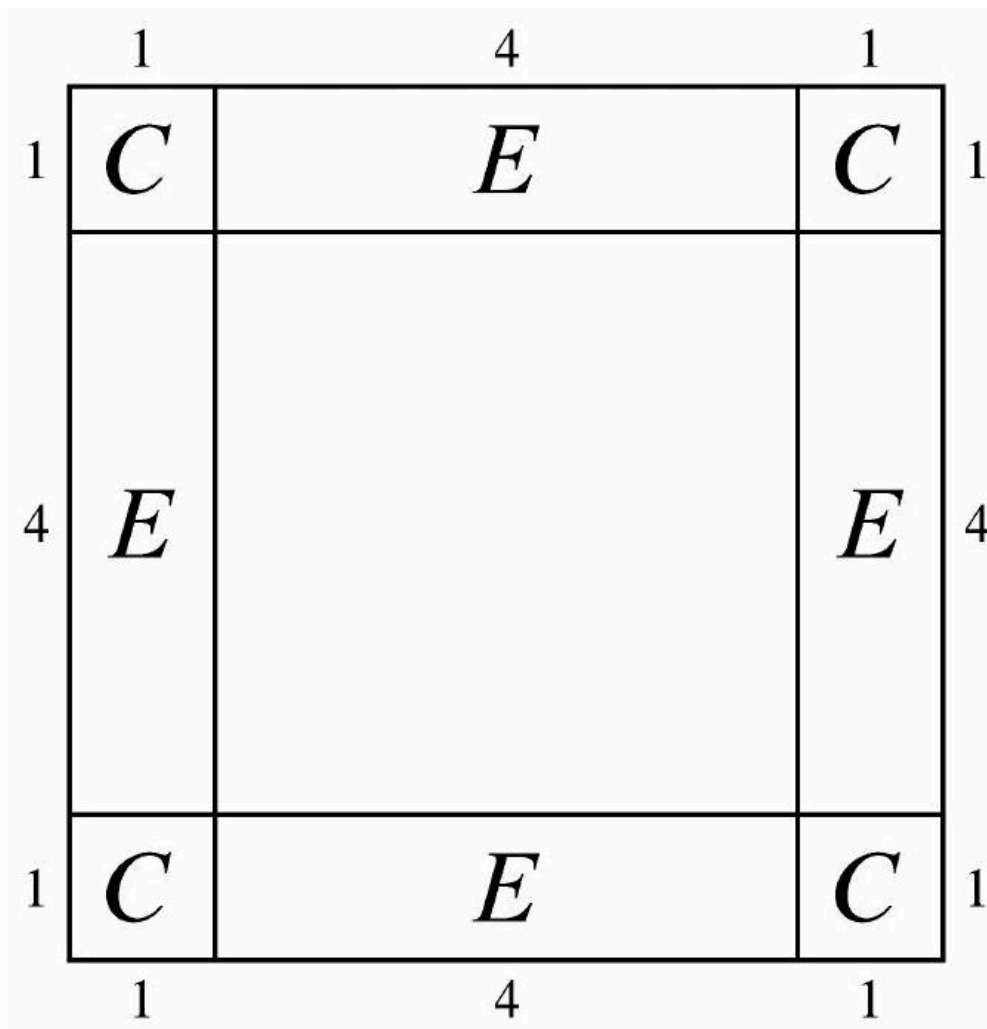
Sonya the frog chooses a point uniformly at random lying within the square $[0, 6] \times [0, 6]$ in the coordinate plane and hops to that point. She then chooses a distance uniformly at random in the interval $[0, 1]$ and a direction uniformly at random from $\{north, east, south, west\}$. All her choices are independent. She now hops the chosen distance in the chosen direction. What is the probability that she lands outside the square?

Answer Choices:

- A. $\frac{1}{6}$
- B. $\frac{1}{12}$
- C. $\frac{1}{4}$
- D. $\frac{1}{10}$
- E. $\frac{1}{9}$

Solution:

In the figure below, Sonya can hop out of the square only if she occupies a region labeled E or C after the first hop.



Let (u, v) be the randomly chosen starting point in the square, and let h be the distance of the second hop, where $0 \leq h \leq 1$. Suppose $(u, v) \in E$, without loss of generality, say in the E -region at the top, so $5 \leq v \leq 6$. Then Sonya will exit the square if and only if the chosen direction is north and $h + v > 6$. Because h is uniformly distributed in $[0, 1]$ and v is uniformly distributed in $[5, 6]$, the probability that $h + v > 6$ is $\frac{1}{2}$. The probability that the initial hop lands in any region labeled E is $\frac{16}{36} = \frac{4}{9}$, and the probability that the chosen direction is serendipitous is $\frac{1}{4}$, for a total probability in this case of

$$\frac{1}{2} \cdot \frac{4}{9} \cdot \frac{1}{4} = \frac{1}{18}.$$

The contribution from the corner regions labeled C is similar, but two of the four possible directions will lead to Sonya leaving the square this time:

$$\frac{4}{36} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{36}.$$

The required probability is

$$\frac{1}{18} + \frac{1}{36} = \text{(B)} \boxed{\frac{1}{12}}$$

Note: For an $n \times n$ square, the probability is $\frac{1}{2n}$.

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